

the averaged attention score to determine which KV embeddings should be retained. However, this alternative approach resulted in performance degradation. Additionally, we observed a significant proportion of H_2 occurrences at the beginning of sentences. This finding suggests that the initial tokens play a substantial role in subsequent generation tasks.

Social Impact. Our work represents an initial effort in designing a KV Cache policy, a realm that has been relatively unexplored and yet is a significant bottleneck in LLMs. The proposed Heavy Hitter Oracle (H_2O) provide a solution to improve the efficiency of LLM generation, which can save energy cost and contribute to green AI. Besides, our approach also serves as a source of inspiration for future advanced algorithm designs. We envision H_2O as a foundational framework that could facilitate further innovation in this area. Moreover, long content generation is an area of growing importance that currently grapples with several efficiency issues. We hope our work that supports the generation of very long sequences will support further research in this direction, particularly in terms of enhancing consistency, devising superior evaluation methods, and establishing robust benchmarks.

Furthermore, another contribution of this study is the formulation of a dynamic submodular framework. We believe that this theoretical framework possesses the potential to be applicable beyond specific domains of interest. For instance, there may exist numerous other dynamic problems where the task involves solving a submodular problem with slight variations at each time.

Limitations. Furthermore, despite the notable advancements in throughput of our H_2O , implementing LLMs for generative inference remains challenging due to the immense parameter count. As a substantial portion of these parameters is encompassed within the MLP blocks, building upon our observations of H_2 occurrences in the MLP blocks, future research efforts can be directed towards leveraging the characteristics of H_2 to devise an offloading policy. Such a policy can potentially enhance the efficiency of LLM inference even further.

C Extended Experiments

In this section, our goal is to demonstrate that H₂O can improve the diversity of generated text (mentioned in Section 5.3), throughput is further improved when combined with quantization (mentioned in Section 5.3), and superior performance to handle extremely long length inputs (up to four middle tokens, mentioned in Section 5.3). Moreover, additional investigations about H₂ are reported, including experiments under zero-shot/one-shot inference regime; H₂O can also enhance the "Top-K" baseline; extra results of H₂ in attention blocks; the emergence of H₂ in MLP blocks as well as its properties.

C.1 Increased Diversity of Generated Text

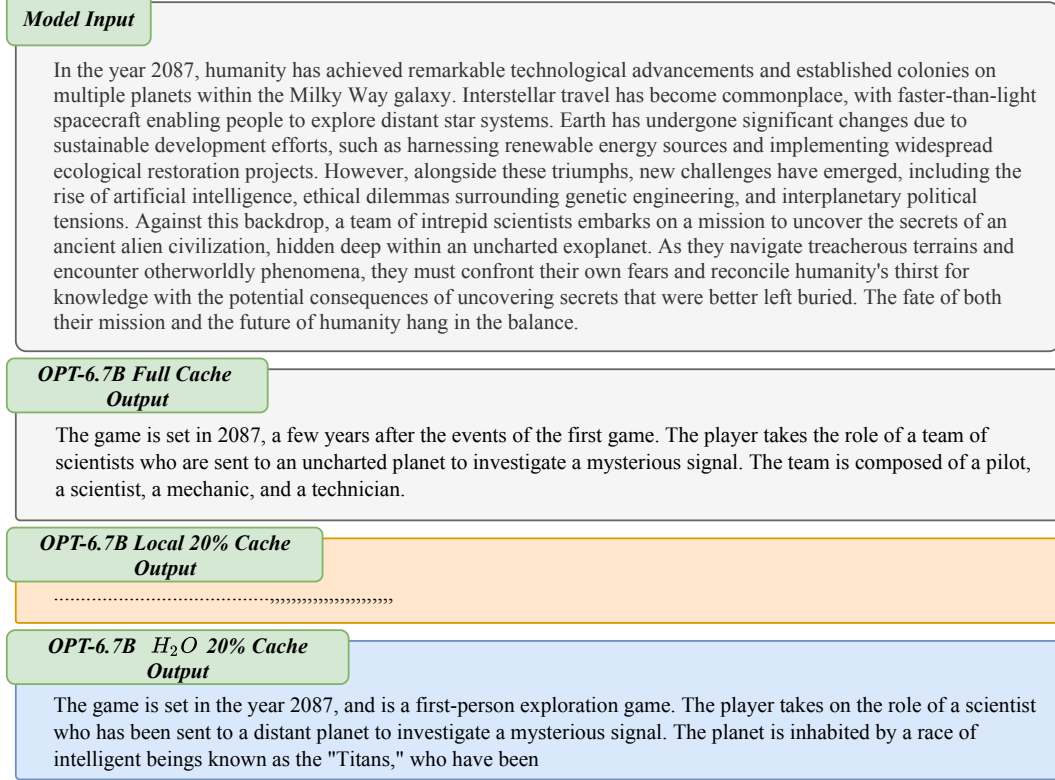


Figure 6: Visualization of one generation example with OPT-6.7B. Results are compared between the baseline model with full cache, our H₂O, and the "Local" strategy that utilizes the most recent KV embeddings.

Given the same prompt text, we visualize the generated text with OPT-6.7B and LLaMA-7B across different methods, including the baseline model with full cache, our H₂O, and the "Local" strategy. Results are reported in Figure 6 and 7. Even with less KV cache budget, our H₂O can generate more diverse content. Specifically, with the OPT-6.7B, the full model generates some redundant works, like "a few years after the events of the first game" while our H₂O describes "the game is a first-person exploration game". As a comparison, when all KV cache budget is assigned to the most recent tokens, the model fails to generate meaningful text and only repeats the word "." and ",". Similar observations can also be drawn from the results of LLaMA-7B, in which the full model repeatedly says "so moving that", "so moved that", and "began to cry" while our H₂O describes both the people and the environment.

Moreover, We conducted a quantitative comparison via diversity metric(Self-BELU [93]). We randomly sample 100 instances from the XSUM dataset, as prompts. Subsequently, we employed the LLaMa-7B to generate text of equal length. The full model reaches a Self-BELU score of 0.0057 while H₂O and the local method achieve 0.0051 and 0.0436, respectively. The comparatively lower Self-BELU score of H₂O indicates the slightly increased diversity of generated text.

Model Input

In a small, bustling cafe nestled in the heart of a vibrant city, a serendipitous event unfolded, leaving a lasting impression on all who witnessed it. As the patrons sat sipping their coffees and engaging in animated conversations, a talented street musician entered the cafe, carrying a weathered guitar and radiating an aura of creativity.

LLaMA-7B Full Cache Output

He began to play, and the patrons were captivated. The musician’s performance was so moving that the patrons began to applaud, and the musician was so moved that he began to cry. The patrons were so moved that they began to cry, and the musician was so

LLaMA-7B Local 20% Cache Output

He ((((((((((((((, []), 1999, 2000, 2001, 1, and, and, and, and, and, and, and, and, and, and, and, and, and,

LLaMA-7B H₂O 20% Cache Output

He began to play, and the room was filled with the sound of his music. The patrons of the cafe were enthralled by the music, and the atmosphere was electric. The cafe was packed with people, all of whom were enjoying the music. The musician was a young

Figure 7: Visualization of one generation example with LLaMA-7B. Results are compared between the baseline model with full cache, our H₂O, and the "Local" strategy that utilizes the most recent KV embeddings.

Table 6: Compatibility with Quantization.

Models	COPA	OpenBookQA	PiQA
Full	85.00	43.20	78.51
H ₂ O	84.00	43.00	78.45
Quant-4bit	84.00	43.28	78.67
H ₂ O w. Quant-4bit	84.00	43.20	78.80

C.2 Throughput Improvement of H₂O Combining with Quantization

Table 6 demonstrates H₂O are capable of quantization and preserve the full model’s performance with even 4-bit quantization. To further explore the compatibility of our H₂O with quantization with respect to throughput improvement, we conduct an additional evaluation with the quantization method implemented in FlexGen (Note that [94] employed a different 4-bit quantization method). The corresponding results are presented in Table 7. Notably, for OPT-6.7B, we observed extra performance enhancements in H₂O when utilizing quantization compared to the vanilla version. This improvement results from the GPU memory freed by weight quantization, which allows for a significant increase in batch size. However, it should be emphasized that the quantization method employed in FlexGen is not implemented most efficiently, resulting in considerable computational overhead. Despite the batch size being enlarged by 20 times, the actual throughput improvement is less than 2 times. Nevertheless, it is important to acknowledge the potential benefits of combining H₂O with quantization, as exemplified by the ability to increase the batch size further. For instance, the implementation of 4-bit quantization could be accelerated by an optimized CUDA kernel.

Table 7: Generation throughput (token/s) with different systems without offloading. We use H₂O-c denotes the H₂O with 4-bit weights compression. In the sequence length row, we use “512 + 32” to denote a prompt length of 512 and a generation length of 32. “OOM” means out-of-memory. The gray text in the bracket denotes the batch size. We run OPT-6.7B on a single T4 GPU.

Seq. length	512+32	512+512	512+1024
Model size	6.7B	6.7B	6.7B
Accelerate	20.4 (2)	15.5 (1)	OOM
DeepSpeed	OOM	OOM	OOM
FlexGen	20.2 (2)	16.8 (1)	16.9 (1)
H ₂ O (20%)	35.1 (4)	51.7 (4)	52.1 (4)
H ₂ O-c (20%)	50.5 (70)	72.5 (52)	62.3 (44)